GED Math Practice Test 8

PART A — QUESTIONS

GED-MATH8-001
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Non-Calculator
Prompt:
A warehouse shelf holds 8 bins. Each bin contains 15 small parts. If 25 parts are removed, how many parts remain on the shelf?
A) 95
B) 105
C) 120
D) 145

GED-MATH8-002
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Non-Calculator
Prompt:
What is the value of $4(2n + 1) - 3n$ when $n = 6$?
A) 22
B) 28
C) 34
D) 52

GED-MATH8-003
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Non-Calculator
Prompt:
A bottle contains 3/5 liter of juice. A student drinks 1/2 of the juice. How many liters of juice does the student drink?
A) 1/10
B) 3/10
C) 1/2
D) 6/5

GED-MATH8-004
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Non-Calculator
Prompt:
Which expression is equivalent to $9 - 2(3x - 5)$?
A) $-6x - 1$
B) $-6x + 14$

C) $6x - 1$
D) $6x + 14$

GED-MATH8-005
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Non-Calculator
Prompt:
A container holds 2.5 gallons of cleaner. How many pints of cleaner does the container hold?
A) 5 pints
B) 10 pints
C) 16 pints
D) 20 pints

GED-MATH8-006
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The formula $A = 4s^2$ gives the area A of one side of a square sign with side length s. If A = 196, what is s?
A) 7
B) 14
C) 28
D) 49

GED-MATH8-007
Section: Math
Type: numeric_entry
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A box has a mass of 18.6 kilograms. Another box has a mass of 7,450 grams. What is the combined mass of the two boxes in kilograms?
Type your answer as a decimal rounded to the nearest hundredth.

GED-MATH8-008
Section: Math
Type: dropdown
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The formula $V = \pi r^2 h$ gives the volume V of a cylinder. To solve for h, divide both sides by [D1], so h = [D2].
Dropdowns:
D1) πr^2 | πh | $r^2 h$
D2) $V/(\pi r^2)$ | $\pi r^2/V$ | $V\pi r^2$

GED-MATH8-009
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A community class has 54 adults and 36 teens. What percent of the class is made up of teens?
A) 36%
B) 40%
C) 54%
D) 60%

GED-MATH8-010
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A value of y satisfies $2.5y - 6 \leq 19$. Which value of y is the greatest possible whole-number solution?
A) 8
B) 9
C) 10
D) 11

GED-MATH8-011
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A baker used 7.5 pounds of flour to make 120 rolls. At the same rate, how many pounds of flour are needed to make 200 rolls?
A) 10.5
B) 11.25
C) 12.5
D) 15

GED-MATH8-012
Section: Math
Type: multi_select
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
Which expressions are equivalent to $15 - 5x$?
A) $5(3 - x)$
B) $-5(x - 3)$
C) $5(x - 3)$
D) $-5x - 15$
E) $3(5 - x)$

GED-MATH8-013
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
The table shows the number of community garden plots by size.

Plot size: Small, Medium, Large
Number of plots: 18, 27, 15

If one plot is selected at random, what is the probability that it is medium or large?
A) 1/4
B) 3/5
C) 7/10
D) 9/10

GED-MATH8-014
Section: Math
Type: numeric_entry
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The formula R = 1.8C + 32 converts a temperature from degrees Celsius C to degrees Fahrenheit R.
If R = 77, what is C?
Type your answer as an integer.

GED-MATH8-015
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A trapezoid has bases of 10 inches and 16 inches and a height of 7 inches. What is the area of the trapezoid?
Use $A = 1/2(b_1 + b_2)h$.
A) 91 square inches
B) 98 square inches
C) 112 square inches
D) 182 square inches

GED-MATH8-016
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The table shows part of a linear relationship.

x: 2, 6, 14, 18
y: 11, 23, 47, 59

Which equation represents the relationship?
A) y = 2x + 7
B) y = 3x + 5
C) y = 4x + 3
D) y = 6x − 1

GED-MATH8-017
Section: Math
Type: multi_select
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
The table shows rainfall amounts for four months.

Month: April, May, June, July
Rainfall: 3.8 in, 5.1 in, 2.4 in, 4.7 in

Which statements are true?
A) The total rainfall was 16.0 inches.
B) The range was 2.7 inches.
C) The mean rainfall was 4.5 inches.
D) May had twice as much rainfall as June.
E) The median rainfall was 4.7 inches.

GED-MATH8-018
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Which inequality represents “four less than the quotient of a number and 3 is greater than 11”?
A) x/3 − 4 > 11
B) 3/x − 4 > 11
C) 4 − x/3 > 11
D) x/(3 − 4) > 11

GED-MATH8-019
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A coat is sold for $144 after a 20% discount. What was the original price of the coat?
A) $115.20
B) $172.80
C) $180.00
D) $288.00

GED-MATH8-020
Section: Math

Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Solve for k: $2(5k - 7) - 3(k + 4) = 44$.
A) 6
B) 8
C) 10
D) 12

GED-MATH8-021
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A rectangular frame has outside dimensions 20 inches by 14 inches. The opening inside the frame is 16 inches by 10 inches. What is the area of the frame only?
A) 80 square inches
B) 120 square inches
C) 160 square inches
D) 280 square inches

GED-MATH8-022
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A line passes through the points $(3, -2)$ and $(9, 16)$. What is the rate of change of the line?
A) 2
B) 3
C) 4
D) 6

GED-MATH8-023
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A solution contains 18 grams of salt in 240 grams of solution. What is the salt concentration as a percent of the solution?
A) 7.5%
B) 8%
C) 12.5%
D) 13.3%

GED-MATH8-024
Section: Math
Type: mcq

Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Function A is represented by $y = -2x + 18$. Function B is represented by the table.

x: 0, 3, 6, 9
y: 20, 11, 2, −7

Which statement is true?
A) Function A decreases faster than Function B.
B) Function B decreases faster than Function A.
C) Both functions have the same rate of change.
D) Function B has a positive rate of change.

GED-MATH8-025
Section: Math
Type: numeric_entry
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A final grade is based on 60% written work, 25% projects, and 15% participation. A student scores 78 on written work, 92 on projects, and 88 on participation. What is the final grade?
Type your answer as a decimal rounded to the nearest tenth.

GED-MATH8-026
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A store sold 72 shirts. Plain shirts cost $12 each, and printed shirts cost $17 each. The total revenue was $1,074. Which system can be used to find p, the number of plain shirts, and r, the number of printed shirts?
A) $p + r = 1{,}074$
$12p + 17r = 72$
B) $p + r = 72$
$12p + 17r = 1{,}074$
C) $12p + 17r = 72$
$p - r = 1{,}074$
D) $17p + 12r = 72$
$p + r = 1{,}074$

GED-MATH8-027
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A circular fountain has a radius of 4.5 feet. Using $\pi \approx 3.14$, what is the area of the fountain's surface?
A) 28.26 square feet

B) 56.52 square feet
C) 63.59 square feet
D) 127.17 square feet

GED-MATH8-028
Section: Math
Type: dropdown
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The equation $6x + 3y = 24$ can be rewritten as y = [D1]x + [D2].
Dropdowns:
D1) −2 | 2 | 6
D2) 8 | 24 | −8

GED-MATH8-029
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A data set has five numbers with a mean of 18. Four of the numbers are 12, 16, 20, and 25. What is the fifth number?
A) 15
B) 17
C) 19
D) 21

GED-MATH8-030
Section: Math
Type: multi_select
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
A function is represented by $f(x) = 2x^2 - 3x + 1$. Which statements are true?
A) $f(0) = 1$
B) $f(1) = 0$
C) $f(2) = 3$
D) $f(-1) = 0$
E) $f(3) = 10$

GED-MATH8-031
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A rectangular tank has a length of 30 centimeters, a width of 18 centimeters, and a height of 20 centimeters. It is filled to 75% of its volume. How many cubic centimeters of water are in the tank?
A) 2,700

B) 5,400
C) 8,100
D) 10,800

GED-MATH8-032
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Solve for m: $8 - 3(m - 5) = 2m + 18$.
A) 1
B) 2
C) 3
D) 5

GED-MATH8-033
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A ladder is 26 feet long and reaches a point 24 feet up a wall. How far is the bottom of the ladder from the wall?
A) 6 feet
B) 8 feet
C) 10 feet
D) 14 feet

GED-MATH8-034
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A quadratic function is defined by $q(x) = x^2 - 10x + 21$. Which value of x makes $q(x) = 0$?
A) 2
B) 3
C) 10
D) 21

GED-MATH8-035
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A pond contains 2,000 fish. A model predicts that the fish population will be 2,000(1.04)^t after t years. What does 1.04 mean in this model?
A) The population increases by 4% each year.
B) The population decreases by 4% each year.

C) The population starts at 104 fish.
D) The population gains 1.04 fish each year.

GED-MATH8-036
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A map scale says 3 centimeters represents 42 kilometers. Two towns are 7.5 centimeters apart on the map. How far apart are the towns?
A) 84 kilometers
B) 96 kilometers
C) 105 kilometers
D) 126 kilometers

GED-MATH8-037
Section: Math
Type: drag_drop
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Match each target with the correct tile.
Tiles:
A) $y = 6x + 9$
B) $y = 9x + 6$
C) $y = 6(x + 9)$
D) $y = 9(x + 6)$
Targets:
T1) Multiply a number by 6, then add 9
T2) Add 9 to a number, then multiply by 6
T3) Add 6 to a number, then multiply by 9

GED-MATH8-038
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A delivery service charges by weight. The cost for 14 pounds is $23.80, and the cost is proportional to weight. What is the cost for 9 pounds?
A) $12.60
B) $15.30
C) $17.20
D) $20.80

GED-MATH8-039
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed

Prompt:
Which ordered pair satisfies the inequality $y \leq -x + 6$?
A) (1, 8)
B) (2, 5)
C) (4, 4)
D) (7, 1)

GED-MATH8-040
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A value is modeled by V = 80(0.75)^t. Which statement best describes the change in V each time t increases by 1?
A) V decreases by 25%.
B) V decreases by 75%.
C) V increases by 25%.
D) V increases by 75%.

GED-MATH8-041
Section: Math
Type: multi_select
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
The table shows the price of two package sizes of almonds.

Package: Small, Large
Weight: 12 oz, 20 oz
Price: $4.20, $6.40

Which statements are true?
A) The small package costs $0.35 per ounce.
B) The large package costs $0.32 per ounce.
C) The small package has the lower unit price.
D) The large package costs $0.40 per ounce.
E) The two packages have the same unit price.

GED-MATH8-042
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
For the function $g(x) = -3x^2 + 2x + 5$, what is $g(-2)$?
A) −21
B) −11
C) 1
D) 13

GED-MATH8-043
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The table shows a relationship.

x: 1, 2, 3, 4
y: 2, 6, 12, 20

Which statement best describes the relationship?
A) It is linear because y increases each time.
B) It is linear because x increases by 1 each time.
C) It is not linear because the changes in y are not constant.
D) It is not linear because the y-values are greater than the x-values.

GED-MATH8-044
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Which expression represents 6 more than half the difference of a number x and 10?
A) $1/2x - 10 + 6$
B) $1/2(x - 10) + 6$
C) $1/2(x + 10) - 6$
D) $6(x - 10)/2$

GED-MATH8-045
Section: Math
Type: numeric_entry
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The formula $A = 1/2h(b_1 + b_2)$ gives the area of a trapezoid. If $A = 84$, $h = 7$, and $b_1 = 9$, what is b_2?
Type your answer as an integer.

GED-MATH8-046
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A garden path is made from 48 identical square stones. Each stone has a side length of 1.5 feet.
What is the total area covered by the stones?
A) 72 square feet
B) 96 square feet
C) 108 square feet
D) 144 square feet

PART B — ANSWER KEY + EXPLANATIONS

GED-MATH8-001 — Correct: A — Correct Answer: 95 — Explanation: The shelf starts with $8 \times 15 = 120$ parts. After 25 are removed, $120 - 25 = 95$ parts remain.

GED-MATH8-002 — Correct: C — Correct Answer: 34 — Explanation: Substitute $n = 6$: $4(2(6) + 1) - 3(6) = 4(13) - 18 = 52 - 18 = 34$.

GED-MATH8-003 — Correct: B — Correct Answer: 3/10 — Explanation: Half of 3/5 is $3/5 \times 1/2 = 3/10$.

GED-MATH8-004 — Correct: B — Correct Answer: $-6x + 14$ — Explanation: Distribute: $9 - 2(3x - 5) = 9 - 6x + 10 = -6x + 19$. Correction: The equivalent expression is $-6x + 19$, which is not listed.

GED-MATH8-005 — Correct: D — Correct Answer: 20 pints — Explanation: There are 8 pints in 1 gallon, so 2.5 gallons $= 2.5 \times 8 = 20$ pints.

GED-MATH8-006 — Correct: A — Correct Answer: 7 — Explanation: $196 = 4s^2$, so $49 = s^2$ and $s = 7$.

GED-MATH8-007 — Correct: 26.05 — Tolerance: exact — Correct Answer: 26.05 — Explanation: 7,450 grams = 7.45 kilograms. The combined mass is $18.6 + 7.45 = 26.05$ kilograms.

GED-MATH8-008 — Correct: D1=πr^2; D2=$V/(\pi r^2)$ — Correct Answer: D1=πr^2; D2=$V/(\pi r^2)$ — Explanation: Divide both sides of $V = \pi r^2 h$ by πr^2 to isolate h.

GED-MATH8-009 — Correct: B — Correct Answer: 40% — Explanation: There are 90 people total, and 36 are teens. Then $36/90 = 0.40$, or 40%.

GED-MATH8-010 — Correct: C — Correct Answer: 10 — Explanation: Solve $2.5y - 6 \leq 19$. Add 6 to get $2.5y \leq 25$, so $y \leq 10$.

GED-MATH8-011 — Correct: C — Correct Answer: 12.5 — Explanation: The flour per roll is $7.5 \div 120 = 0.0625$ pound. For 200 rolls, $200 \times 0.0625 = 12.5$ pounds.

GED-MATH8-012 — Correct: A,B — Correct Answer: $5(3 - x)$ | $-5(x - 3)$ — Explanation: Both expressions simplify to $15 - 5x$.

GED-MATH8-013 — Correct: C — Correct Answer: 7/10 — Explanation: There are 60 plots total. Medium or large plots total $27 + 15 = 42$. The probability is $42/60 = 7/10$.

GED-MATH8-014 — Correct: 25 — Tolerance: exact — Correct Answer: 25 — Explanation: $77 = 1.8C + 32$. Subtract 32 to get $45 = 1.8C$, so $C = 25$.

GED-MATH8-015 — Correct: A — Correct Answer: 91 square inches — Explanation: $A = 1/2(10 + 16)(7) = 1/2(26)(7) = 91$ square inches.

GED-MATH8-016 — Correct: B — Correct Answer: $y = 3x + 5$ — Explanation: When x increases by 4, y increases by 12, so the rate is 3. Substituting $x = 2$ and $y = 11$ gives $y = 3x + 5$.

GED-MATH8-017 — Correct: A,B — Correct Answer: The total rainfall was 16.0 inches. | The range was 2.7 inches. — Explanation: Total rainfall is $3.8 + 5.1 + 2.4 + 4.7 = 16.0$ inches. The range is $5.1 - 2.4 = 2.7$ inches.

GED-MATH8-018 — Correct: A — Correct Answer: $x/3 - 4 > 11$ — Explanation: "The quotient of a number and 3" is $x/3$, and "four less than" that is $x/3 - 4$.

GED-MATH8-019 — Correct: C — Correct Answer: \$180.00 — Explanation: After a 20% discount, the sale price is 80% of the original price. So $144 \div 0.80 = 180$.

GED-MATH8-020 — Correct: C — Correct Answer: 10 — Explanation: $2(5k - 7) - 3(k + 4) = 10k - 14 - 3k - 12 = 7k - 26$. Then $7k - 26 = 44$, so $k = 10$.

GED-MATH8-021 — Correct: B — Correct Answer: 120 square inches — Explanation: Outside area is $20 \times 14 = 280$. Opening area is $16 \times 10 = 160$. Frame area is $280 - 160 = 120$ square inches.

GED-MATH8-022 — Correct: B — Correct Answer: 3 — Explanation: Rate of change $= (16 - (-2))/(9 - 3) = 18/6 = 3$.

GED-MATH8-023 — Correct: A — Correct Answer: 7.5% — Explanation: $18 \div 240 = 0.075$, or 7.5%.

GED-MATH8-024 — Correct: B — Correct Answer: Function B decreases faster than Function A. — Explanation: Function A has rate -2. Function B has rate $(11 - 20)/(3 - 0) = -3$, so Function B decreases faster.

GED-MATH8-025 — Correct: 83.0 — Tolerance: exact — Correct Answer: 83.0 — Explanation: Final grade $= 0.60(78) + 0.25(92) + 0.15(88) = 46.8 + 23.0 + 13.2 = 83.0$.

GED-MATH8-026 — Correct: B — Correct Answer: $p + r = 72$
$12p + 17r = 1{,}074$ — Explanation: The first equation represents the number of shirts, and the second represents total revenue.

GED-MATH8-027 — Correct: C — Correct Answer: 63.59 square feet — Explanation: Area $= \pi r^2 = 3.14(4.5^2) = 3.14(20.25) = 63.585$, which rounds to 63.59.

GED-MATH8-028 — Correct: D1=−2; D2=8 — Correct Answer: D1=−2; D2=8 — Explanation: $6x + 3y = 24$ gives $3y = -6x + 24$, so $y = -2x + 8$.

GED-MATH8-029 — Correct: B — Correct Answer: 17 — Explanation: Five numbers with mean 18 must total 90. The four given numbers total 73, so the fifth number is $90 - 73 = 17$.

GED-MATH8-030 — Correct: A,B — Correct Answer: $f(0) = 1$ | $f(1) = 0$ — Explanation: $f(0) = 1$, and $f(1) = 2 - 3 + 1 = 0$.

GED-MATH8-031 — Correct: C — Correct Answer: 8,100 — Explanation: Full volume is $30 \times 18 \times 20 = 10{,}800$ cubic centimeters. Seventy-five percent of that is $0.75 \times 10{,}800 = 8{,}100$.

GED-MATH8-032 — Correct: A — Correct Answer: 1 — Explanation: $8 - 3(m - 5) = 8 - 3m + 15 = 23 - 3m$. Then $23 - 3m = 2m + 18$, so $5 = 5m$ and $m = 1$.

GED-MATH8-033 — Correct: C — Correct Answer: 10 feet — Explanation: The distance from the wall is $\sqrt{(26^2 - 24^2)} = \sqrt{(676 - 576)} = \sqrt{100} = 10$.

GED-MATH8-034 — Correct: B — Correct Answer: 3 — Explanation: $q(x) = x^2 - 10x + 21$ factors as $(x - 3)(x - 7)$, so $x = 3$ makes $q(x) = 0$.

GED-MATH8-035 — Correct: A — Correct Answer: The population increases by 4% each year. — Explanation: Multiplying by 1.04 each year means the population keeps 100% and increases by 4%.

GED-MATH8-036 — Correct: C — Correct Answer: 105 kilometers — Explanation: The scale is $42 \div 3 = 14$ kilometers per centimeter. Then $7.5 \times 14 = 105$ kilometers.

GED-MATH8-037 — Correct: T1=A; T2=C; T3=D — Correct Answer: T1=$y = 6x + 9$; T2=$y = 6(x + 9)$; T3=$y = 9(x + 6)$ — Explanation: Parentheses show which addition happens before multiplication.

GED-MATH8-038 — Correct: B — Correct Answer: \$15.30 — Explanation: The unit price is $23.80 \div 14 = \$1.70$ per pound. For 9 pounds, the cost is $9 \times 1.70 = \$15.30$.

GED-MATH8-039 — Correct: B — Correct Answer: (2, 5) — Explanation: Substitute (2, 5): $-2 + 6 = 4$, and $5 \leq 4$ is false. Correction: No listed ordered pair satisfies the inequality.

GED-MATH8-040 — Correct: A — Correct Answer: V decreases by 25%. — Explanation: Multiplying by 0.75 means 75% of the value remains, so the decrease is 25%.

GED-MATH8-041 — Correct: A,B — Correct Answer: The small package costs \$0.35 per ounce. | The large package costs \$0.32 per ounce. — Explanation: Small unit price: $4.20 \div 12 = \$0.35$ per ounce. Large unit price: $6.40 \div 20 = \$0.32$ per ounce.

GED-MATH8-042 — Correct: B — Correct Answer: −11 — Explanation: $g(-2) = -3(-2)^2 + 2(-2) + 5 = -12 - 4 + 5 = -11$.

GED-MATH8-043 — Correct: C — Correct Answer: It is not linear because the changes in y are not constant. — Explanation: The y-values increase by 4, then 6, then 8, so the rate of change is not constant.

GED-MATH8-044 — Correct: B — Correct Answer: $1/2(x - 10) + 6$ — Explanation: “The difference of x and 10” is $x - 10$. Half that difference is $1/2(x - 10)$, and 6 more is $1/2(x - 10) + 6$.

GED-MATH8-045 — Correct: 15 — Tolerance: exact — Correct Answer: 15 — Explanation: $84 = 1/2(7)(9 + b_2)$, so $84 = 3.5(9 + b_2)$. Then $24 = 9 + b_2$, so $b_2 = 15$.

GED-MATH8-046 — Correct: C — Correct Answer: 108 square feet — Explanation: Each square stone has area $1.5 \times 1.5 = 2.25$ square feet. The total area is $48 \times 2.25 = 108$ square feet.